Simplified Economic Evaluation Model of a Fictional Mining Project

A Simplified Discounted Cash Flow Analysis

Academic work for educational purposes

Lucas Lokumu

Disclaimer: this report concerns an entirely fictional mining project, built for educational purposes. It does not describe any real deposit, company, or feasibility study.

1. Introduction

Mining projects are among the longest, most costly, and most uncertain investments that exist. Before a single tonne of ore is sold, a company must commit substantial spending: building facilities, purchasing equipment, preparing the site. Only after several years of construction does the mine become productive and start generating revenue. Between the moment the money is spent and the moment it is recovered, several years may pass, during which the economic environment, market prices, or production costs can evolve in unpredictable ways.

The economic evaluation of a mining project addresses precisely this difficulty: it makes it possible to compare, on a common basis, the costs incurred today and the revenues expected in the future, in order to assess whether the project is capable of creating value. This comparison relies on standard financial tools, but applying them to the mining sector requires an understanding of certain specificities of this type of investment, in particular the existence of a construction phase with no revenue and a closure phase at the end of the mine's life.

This report presents a simplified economic evaluation model of a fictional mining project, built from an Excel file structured into several worksheets. The aim is not to produce a feasibility study, but to understand, step by step, the mechanisms that make it possible to move from starting assumptions (production, price, costs) to the financial indicators used to judge the merits of an investment project. It is important to state, from this introduction onward, that all the data used are fictional and that the model has a strictly educational purpose: it does not correspond to any real mining project and cannot replace a mining feasibility study conducted by professionals in the sector.

2. Simplified Presentation of a Mining Project

Before going into the details of the calculations, it is useful to understand, in broad terms, how the life of a mining project unfolds. Three main phases are generally distinguished, each corresponding to a different economic logic.

2.1 The construction phase

At the very start of the project, the mine does not yet exist in a productive form. The company must invest in building the necessary infrastructure: access roads, buildings, a processing plant, extraction and transport equipment. This phase generates no revenue at all, since nothing is yet produced or sold; it translates only into substantial cash outflows. In the model presented here, this phase lasts two years.

2.2 The production phase

Once the facilities are completed, the mine enters operation. Ore is extracted, processed, and sold, which generates revenue. In return, the company must bear ongoing operating costs:

wages, energy, equipment maintenance, transport, ore processing. It is during this phase, generally the longest, that the project is expected to repay the initial investment and generate a surplus of value.

2.3 The closure phase

When the deposit is exhausted or the planned life of the mine comes to an end, the site must be closed. This closure is not free: facilities must be secured, certain equipment dismantled, and the environment affected by the operation restored. This closure cost represents a final cash outflow, which reduces the result of the project's last year.

It should be noted that, in this model, the “mining product” is deliberately simplified. The analysis relies solely on annual production expressed in tonnes, sold at a given price. The model does not attempt to represent the geology of the deposit, the ore grade (that is, the concentration of the useful substance in the extracted rock), or the metallurgical recovery rate (the share of ore actually recovered during processing). These elements, essential in an actual technical study, are set aside here in order to focus on the financial logic of the project.

3. Definitions of Mining and Economic Concepts

This section gathers, in the form of short definitions, the main technical concepts used throughout the rest of the report. Its purpose is to allow the document to be read independently, without prior knowledge of finance or mining economics.

Mining project — An investment aimed at extracting a mineral resource from the subsoil and selling it in a marketable form.

Annual production — The quantity of ore produced and sold each year during the mine's operating phase.

Selling price — The price at which each tonne produced is sold on the market. It directly determines the level of revenue generated by the project.

Revenue — The total amount obtained from the sale of production. It is calculated as follows: annual production multiplied by the selling price.

OPEX — Short for “operating expenditures,” that is, operating costs. These are the expenses required for the day-to-day running of the mine once it is in production: wages, energy, maintenance, transport, ore processing. In the model, OPEX is expressed per tonne produced.

CAPEX — Short for “capital expenditures,” that is, initial investment costs. These are the costs required to build the mine before it begins producing: equipment, the processing plant, infrastructure.

Taxes / royalties — Levies paid to public authorities or to holders of resource rights. In the model, they are simplified as a single rate applied to the pre-tax result.

Closure cost — The cost paid at the end of the project to close the site, secure the facilities, and restore the environment affected by the operation.

Cash flow (net cash flow) — The amount of money actually generated or spent by the project during a given year. It takes into account revenue, operating costs, investment, taxes, and, where applicable, the closure cost.

Cumulative cash flow — The running total of cash flows since the start of the project. It shows the point at which the project has recovered its initial investment.

Discounting — A method used to bring future cash flows back to their equivalent value today. A dollar received in ten years is worth less than a dollar received today, because the future is uncertain and that money could, in the meantime, be invested elsewhere and earn a return.

Discount rate — The rate used to convert future cash flows into present value. It reflects the passage of time, the risk associated with the project, and the opportunity cost of capital, that is, the return that could have been earned by investing the money elsewhere.

NPV (net present value) — An indicator that measures the value created today by the project, once all future cash flows have been discounted and added together. A positive NPV means that the project creates value under the assumptions used; a negative NPV means that it destroys value.

IRR (internal rate of return) — The discount rate that exactly sets the NPV to zero. It provides a measure of the project's internal return. If the IRR is higher than the discount rate used, the project is considered profitable according to this criterion.

Payback period — The point at which the cumulative cash flow turns positive, that is, the point at which the project has fully recovered its initial investment.

Scenario — A coherent set of assumptions used to represent a possible future for the project. Three scenarios are used in this report: pessimistic, central, and optimistic.

4. Presentation of the Project's and Central Scenario's Assumptions

The model relies on a set of starting assumptions, summarized in the table below. These assumptions are broken down into three scenarios — pessimistic, central, and optimistic — which will be compared in detail in section 7. The table below presents all of these assumptions; the paragraphs that follow return more specifically to the central scenario, which serves as the reference for presenting the calculation method.

Table 1 — Project assumptions

Assumption	Central	Pessimistic	Optimistic	Unit
Mine lifespan	12	12	12	years
Construction years	2	2	2	years
Annual production	100,000	100,000	100,000	tonnes
Selling price	1,500	1,125	1,875	$/tonne
OPEX per tonne	700	805	665	$/tonne
Initial CAPEX	300,000,000	360,000,000	270,000,000	$
Discount rate	8%	8%	8%	%
Taxes / royalties	10%	10%	10%	%
Closure cost	30,000,000	30,000,000	30,000,000	$

The central scenario

In the central scenario, the mine produces for twelve years, after two years of construction. During these first two years, the project generates no revenue but bears substantial investment spending. Annual production is set at 100,000 tonnes, sold at a central price of $1,500 per tonne. The operating cost (OPEX) amounts to $700 per tonne produced. The initial CAPEX, which totals $300 million, is split evenly over the two years of construction, that is, -$150 million in year 0 and -$150 million in year 1. The discount rate used is 8%, and a closure cost of $30 million is paid in the project's final year, in addition to normal operating costs.

Reminder: all of these figures are fictional and were chosen solely for educational illustration purposes. They do not describe either a real deposit or an actual market environment.

5. Method for Calculating Cash Flows

The model is built year by year, from year 0 to the end of the mine's lifespan. This construction follows the three-phase breakdown presented in section 2: years 0 and 1 correspond to construction, years 2 to 13 correspond to production, and the last production year (year 13) additionally includes the site closure cost.

Table 2 — Formulas used in the model

Calculated item	Formula	Simple explanation
Revenue	Annual production × Selling price	Amount generated by the sale of production
Annual OPEX	Annual production × OPEX per tonne	Annual cost of running the mine
Pre-tax result	Revenue − OPEX	Margin before taxes and royalties
Taxes / royalties	Pre-tax result × 10%	Simplified levy applied to a positive result
Net cash flow	Revenue − OPEX − CAPEX − Taxes − Closure cost	Net amount of money generated or spent each year
Discounted cash flow	Net cash flow / (1 + discount rate)^year	Present value of the annual cash flow
NPV	Sum of discounted cash flows	Total value created today
IRR	Rate that sets the NPV to zero	Internal return of the project
Payback period	Point at which the cumulative cash flow turns positive	Time needed to recover the investment

Example of a normal production year (central scenario)

To illustrate the method concretely, it is useful to detail the calculation of the cash flow for a normal production year in the central scenario, that is, a year that includes neither construction spending nor a closure cost.

- Annual production = 100,000 tonnes
- Selling price = $1,500/tonne
- Revenue = 100,000 × 1,500 = $150,000,000
- OPEX = 100,000 × 700 = $70,000,000
- Pre-tax result = 150,000,000 − 70,000,000 = $80,000,000
- Taxes / royalties = 80,000,000 × 10% = $8,000,000
- Normal annual net cash flow = 150,000,000 − 70,000,000 − 8,000,000 = $72,000,000

The project's final year (year 13) is calculated in the same way, except that the $30 million closure cost must also be subtracted. The net cash flow for this final year is therefore noticeably lower than that of a normal production year, since it must absorb this final charge.

Detailed year-by-year cash flow tables for the three scenarios (central, pessimistic, and optimistic) are presented in the appendix. They make it possible to verify the model's full construction, from the construction years through to the last production year, including the closure cost.

6. Results of the Central Scenario

Once the annual cash flows have been calculated and discounted, the four summary indicators shown in the following table are obtained.

Table 3 — Results of the central scenario

Indicator	Result	Interpretation
NPV	$202.5 M	The project creates value
IRR	19%	Return higher than the 8% discount rate
Payback period	5.17 years	Investment recovered during year 6
Average annual cash flow	$72.0 M	Average cash generated during normal production

The positive NPV of $202.5 million means that, under the central scenario's assumptions, the discounted future cash flows largely cover the initial investment, operating costs, taxes, and closure cost. Under these conditions, the project therefore creates value.

The IRR of 19% is higher than the 8% discount rate used in the model. This means that the project's internal return exceeds the minimum return required for the investment to be considered acceptable.

The payback period is 5.17 years. In practical terms, the cumulative cash flow turns positive between year 5 and year 6: at the end of year 5, $12 million is still needed to fully recover the investment, and year 6 generates a cash flow of $72 million. The exact calculation of the period is therefore as follows: 5 + 12 / 72 = 5.17 years.

Finally, the average annual cash flow during normal production stands at $72.0 million, which corresponds to the cash generated each year once the mine is in operation, excluding the closure year.

7. Scenario Analysis

The central scenario gives a useful picture of the project, but it is not, on its own, sufficient to judge its merits. Mining projects are indeed exposed to numerous uncertainties: changes in the selling price on international markets, variations in operating costs, cost overruns during construction, market conditions, technical risks. This is why the model incorporates two alternative scenarios, pessimistic and optimistic, built from consistent variations in the main assumptions.

Table 4 — Scenario assumptions

Scenario	Selling price	OPEX per tonne	Initial CAPEX
Pessimistic	$1,125/tonne	$805/tonne	$360,000,000
Central	$1,500/tonne	$700/tonne	$300,000,000
Optimistic	$1,875/tonne	$665/tonne	$270,000,000

The pessimistic scenario combines a lower selling price, higher OPEX, and higher CAPEX, corresponding to an environment that is unfavorable to the project across all of its key parameters. The central scenario reuses the base assumptions presented in section 4. The optimistic scenario, conversely, combines a higher selling price, lower OPEX, and lower CAPEX, corresponding to a particularly favorable environment.

Table 5 — Compared results across scenarios

Scenario	NPV	IRR	Payback	Average annual cash flow	Reading
Pessimistic	-$156.7 M	-2%	Not recovered	$28.8 M	Not profitable
Central	$202.5 M	19%	5.17 years	$72.0 M	Profitable
Optimistic	$488.9 M	33%	4.48 years	$108.9 M	Very attractive

In the pessimistic scenario, the NPV becomes negative and the IRR falls below the 8% discount rate. The project therefore destroys value under these assumptions, and the initial investment is never recovered over the mine's lifespan, hence the “not recovered” label for the payback period.

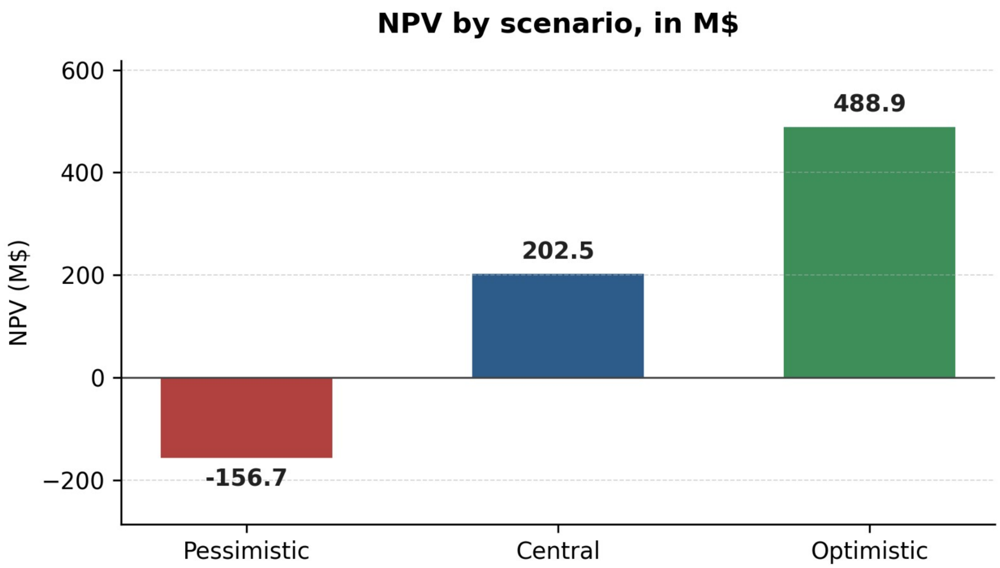
In the central scenario, as detailed in section 6, the project is profitable: the NPV is positive, the IRR exceeds the discount rate, and the investment is recovered before the end of the mine's lifespan.

In the optimistic scenario, the project becomes very attractive, with an NPV of $488.9 million, an IRR of 33%, and a payback period reduced to 4.48 years. This comparison shows that the project is highly sensitive to the assumptions made regarding selling price, CAPEX, and OPEX: a relatively moderate shift in these three parameters is enough to move the project from a value-destroying situation to one of substantial value creation.

8. Charts

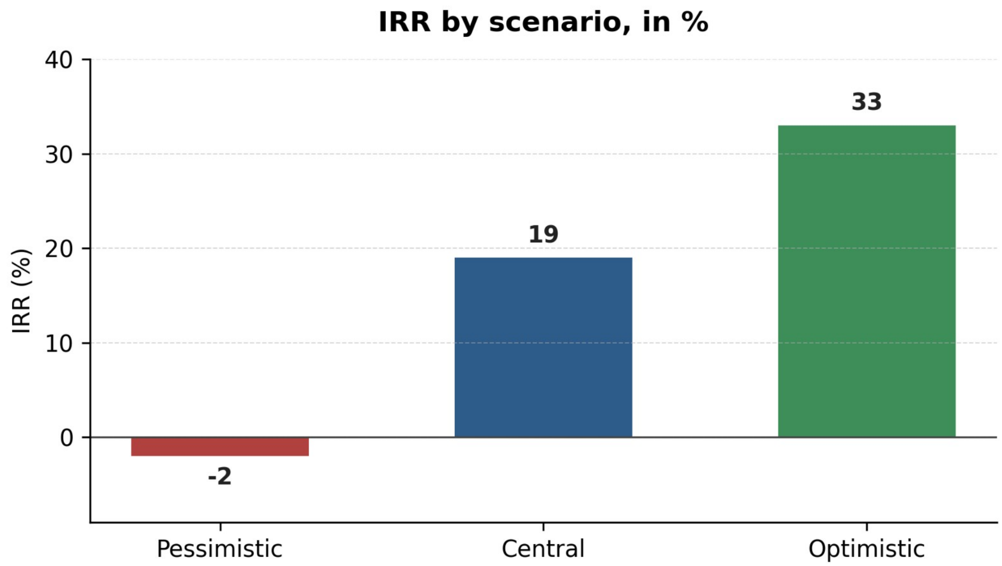
The three charts below reproduce the compared results shown in table 5 and make it possible to visualize, in addition to the figures, the scale of the differences between scenarios. Each chart is accompanied by a short interpretation that draws out the lessons useful for analyzing the project.

8.1 Chart 1 — NPV by scenario, in M$



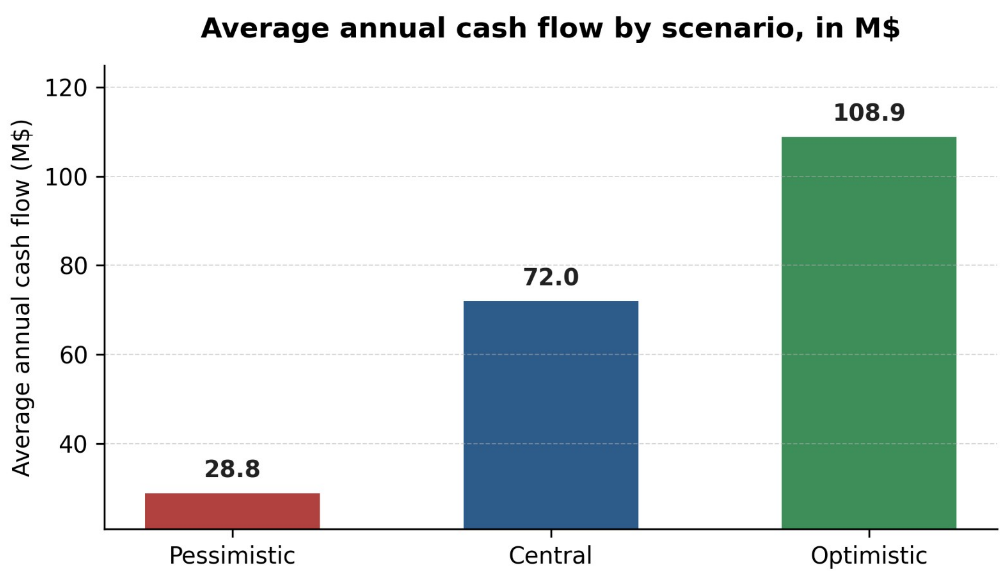
This first chart shows that the NPV is negative in the pessimistic scenario, while it becomes positive in the central scenario and strongly positive in the optimistic scenario. The gap between the three bars illustrates the project's sensitivity to the economic assumptions used: a drop in the selling price combined with an increase in CAPEX and OPEX is, on its own, enough to make the project unprofitable.

8.2 Chart 2 — IRR by scenario, in %



This second chart compares the project's internal return across the three scenarios. In the pessimistic scenario, the IRR is negative, and therefore below the 8% discount rate, which confirms that the project is not viable under these conditions. In the central scenario, the IRR reaches 19%, a level above the discount rate. In the optimistic scenario, it reaches 33%, reflecting a high internal return.

8.3 Chart 3 — Average annual cash flow by scenario, in M$



This third chart shows that the project's ability to generate cash each year, during normal production, varies considerably depending on the assumptions used. In the pessimistic scenario, the average annual cash flow drops to $28.8 M, compared with $72.0 M in the central scenario and $108.9 M in the optimistic scenario. This variation illustrates the direct impact of the selling price and operating costs on the project's ongoing operating profitability, independently even of the question of recovering the initial investment.

9. Limitations of the Model

The model presented in this report relies on numerous simplifications, which need to be made explicit so as not to overinterpret the results obtained. It should first be recalled that the project described is entirely fictional and that the data used does not come from any official technical report; it was chosen solely to illustrate how an economic evaluation model works.

Several simplifying assumptions deserve to be highlighted. The model does not account for inflation, and both the selling price and annual production are assumed constant over the mine's entire lifespan, whereas in reality these two parameters generally vary from one year to the next. The taxation itself is highly simplified, since it amounts to a single rate applied to the pre-tax result, without regard for actual tax rules, which are often more complex. Debt and the associated interest from any external financing are not modeled, nor are accounting depreciation charges, which nonetheless play an important role in the actual calculation of taxes.

From a technical standpoint, the model does not take into account ore grade, that is, the concentration of the useful substance in the extracted rock, nor the metallurgical recovery rate, which measures the share of ore actually recovered during processing. Both of these elements nonetheless strongly influence a mining project's actual revenue. Likewise, detailed environmental costs are not monetized beyond the flat closure cost used, and social, political, and regulatory risks, which are very real in the mining sector, are not explicitly built into the model.

For all of these reasons, this model does not make it possible to conclude on the actual profitability of a mining project. It constitutes a first educational approach, which makes it possible to understand the main economic mechanisms of an extraction project, but which in no way replaces a preliminary economic assessment (PEA), a pre-feasibility study (PFS), or an actual mining feasibility study, which draw on much more detailed geological, technical, and financial data.

10. Conclusion

This work has shown how a simplified economic evaluation can be built from a relatively small set of technical and financial assumptions: annual production, a selling price, operating costs, an initial investment, a discount rate, and a closure cost. From this limited set of data alone, it is possible to calculate annual cash flows, discount them, and derive summary indicators such as the NPV, the IRR, and the payback period.

In the central scenario, the model indicates positive profitability, with an NPV of $202.5 million and an IRR of 19%. At first glance, this result might suggest that the project is solid. However, the scenario analysis showed that this conclusion is fragile: in the pessimistic scenario, which combines a lower selling price, higher operating costs, and higher CAPEX, the project becomes unprofitable. This sensitivity confirms that it would be unwise to judge a mining project solely on the basis of a central assumption, without taking into account the uncertainties weighing on prices and costs.

Beyond the numerical results, this work is above all a useful learning exercise for understanding the logic of investment analysis as applied to natural resources, mining projects, and, more broadly, critical metals, whose economic and strategic importance continues to grow. It is finally worth recalling, one last time, that this is an educational exercise based on fictional data, and not a mining feasibility study in the proper sense.

Appendix A — Cash Flow Summary, Central Scenario

The table below details, year by year, the construction of the cash flow for the central scenario, from the two construction years through to the last production year, including the closure cost.

Year	Phase	Prod.	Price	Revenue	OPEX	CAPEX	Taxes/ Royalties	Closure	Net cash flow	Discounted cash flow	Cumulative cash flow
0	Construction	0	0	0	0	-150,000,000	0	0	-150,000,000	-150,000,000	-150,000,000
1	Construction	0	0	0	0	-150,000,000	0	0	-150,000,000	-138,888,889	-300,000,000
2	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	61,728,395	-228,000,000
3	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	57,155,921	-156,000,000
4	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	52,922,149	-84,000,000
5	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	49,001,990	-12,000,000
6	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	45,372,213	60,000,000
7	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	42,011,308	132,000,000
8	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	38,899,360	204,000,000
9	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	36,017,926	276,000,000
10	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	33,349,931	348,000,000
11	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	30,879,566	420,000,000
12	Production	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	0	72,000,000	28,592,191	492,000,000
13	Production + closure	100,000	1,500	150,000,000	-70,000,000	0	-8,000,000	-30,000,000	42,000,000	15,443,313	534,000,000

Appendix B — Cash Flow Summary, Pessimistic Scenario

The table below details, year by year, the construction of the cash flow for the pessimistic scenario, from the two construction years through to the last production year, including the closure cost.

Year	Phase	Prod.	Price	Revenue	OPEX	CAPEX	Taxes/ Royalties	Closure	Net cash flow	Discounted cash flow	Cumulative cash flow
0	Construction	0	0	0	0	-180,000,000	0	0	-180,000,000	-180,000,000	-180,000,000
1	Construction	0	0	0	0	-180,000,000	0	0	-180,000,000	-166,666,667	-360,000,000
2	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	24,691,358	-331,200,000
3	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	22,862,369	-302,400,000
4	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	21,168,860	-273,600,000
5	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	19,600,796	-244,800,000
6	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	18,148,885	-216,000,000
7	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	16,804,523	-187,200,000
8	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	15,559,744	-158,400,000
9	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	14,407,170	-129,600,000
10	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	13,339,972	-100,800,000
11	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	12,351,826	-72,000,000
12	Production	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	0	28,800,000	11,436,876	-43,200,000
13	Production + closure	100,000	1,125	112,500,000	-80,500,000	0	-3,200,000	-30,000,000	-1,200,000	-441,238	-44,400,000

Appendix C — Cash Flow Summary, Optimistic Scenario

The table below details, year by year, the construction of the cash flow for the optimistic scenario, from the two construction years through to the last production year, including the closure cost.

Year	Phase	Prod.	Price	Revenue	OPEX	CAPEX	Taxes/ Royalties	Closure	Net cash flow	Discounted cash flow	Cumulative cash flow
0	Construction	0	0	0	0	-135,000,000	0	0	-135,000,000	-135,000,000	-135,000,000
1	Construction	0	0	0	0	-135,000,000	0	0	-135,000,000	-125,000,000	-270,000,000
2	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	93,364,198	-161,100,000
3	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	86,448,331	-52,200,000
4	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	80,044,751	56,700,000
5	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	74,115,510	165,600,000
6	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	68,625,472	274,500,000
7	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	63,542,104	383,400,000
8	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	58,835,282	492,300,000
9	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	54,477,113	601,200,000
10	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	50,441,771	710,100,000
11	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	46,705,343	819,000,000
12	Production	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	0	108,900,000	43,245,688	927,900,000
13	Production + closure	100,000	1,875	187,500,000	-66,500,000	0	-12,100,000	-30,000,000	78,900,000	29,011,366	1,006,800,000

Final reminder: all data, assumptions, and results presented in this report are fictional and intended solely for educational purposes. This document does not describe any real mining project and does not, under any circumstances, constitute a mining feasibility study.